

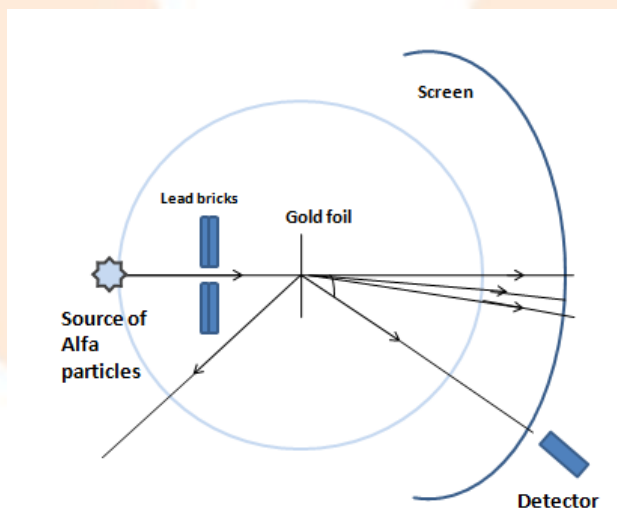
Revision Notes for Class 12 Physics

Chapter 12 – Atoms

Rutherford's Alpha Scattering Experiment:

Following are the observations made through this experiment:

1. Most of α -particles were seen to pass through the gold foil without any appreciable deflection.
2. The various α particles, on passing through the gold foil, undergo different amounts of deflections. A large number of α particles caused fairly large deflections.
3. A very small number of α -particles (about 1 in 8000) practically retraced their paths or suffered deflection of nearly 180° .
4. The plot between the total number of α - particles $N(\theta)$ scattered through angle θ and the scattering angle θ was found to be as shown in graph:



These experimental observations led Rutherford to the following inferences:

1. As most of the α particles passed without any deviation, the atom had a lot of empty space in it.
2. As fast and heavy α particles could be deflected even through 180° , the whole of the positive charge and practically the entire mass of the atom were confirmed to extremely small central cores. These were called nuclei. Because 1 in about 8000 α particles get deflected through 180° , the size of the nucleus was assumed to be about $1/10000th$ of the size of the atom.

8.1 Rutherford's Atom Model

On the basis of the results of α scattering experiment, Rutherford suggested the following picture of an atom:

1. Atoms can be regarded as spheres of diameters $10^{-10}m$ but whole of the positive charge and almost the entire mass of these atoms are concentrated in small central cores called nuclei having diameters of about $10^{-14}m$.
2. The nucleus is neighbored by electrons. In other words, the electrons are distributed over the remaining part of the atom leaving plenty of empty space in the atom.

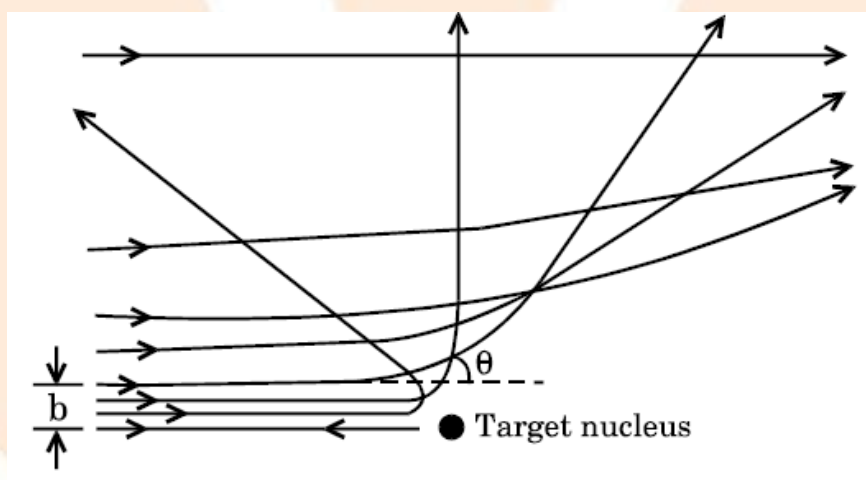
Drawbacks of Rutherford's Atom Model:

I. When the electrons revolve around the nucleus, they get continuously accelerated towards the centre of the nucleus. According to Lorentz, an accelerated charged particle must radiate energy continuously. Thus, in the atom, a revolving electron must continuously emit energy and hence the radius of its path must go on decreasing and finally, it must fall into the nucleus. However, electrons revolve around the nucleus without falling into it. Clearly, Rutherford's atom model couldn't explain the stability of the atom.

II. Suppose if Rutherford's atom model is true, the electron could revolve in orbits of all possible radii and thus it should emit a continuous energy spectrum. But, atoms like hydrogen possess a line spectrum.

Alpha-particle Trajectory:

- The path of an α -particle depends on the impact parameter, which is the distance from the particle's starting point to the centre of the nucleus.
- In a beam of α -particles, each particle has a different impact parameter, causing them to scatter in various directions. They all have nearly the same energy.
- An α -particle that comes very close to the nucleus (small impact parameter) scatters a lot. If it hits the nucleus head-on (very small impact parameter), it bounces back ($\theta \approx 180^\circ$).
- If the impact parameter is large, the α -particle barely changes direction ($\theta \approx 0^\circ$).
- The fact that few α -particles bounce back means that head-on collisions are rare, showing that most of the atom's mass and positive charge is concentrated in a small area. This helps determine the size of the nucleus.



Electron Orbits (Rutherford Model):

1. Rutherford's Nuclear Model:

The atom is modelled as a small, dense, positively charged nucleus with electrons revolving around it in fixed orbits.

The force of attraction between the nucleus and the electrons provides the necessary centripetal force to keep the electrons in stable orbits.

2. Centripetal Force:

The electrostatic force (F_e) between the electron and the nucleus equals the centripetal force (F_c):

$$F_e = F_c = \frac{1}{4\pi\epsilon_0} \frac{e^2}{r^2} = \frac{mv^2}{r}$$

where:

ϵ_0 = permittivity of free space

e = electron charge

r = orbit radius

m = electron mass

v = velocity of the electron

3. Relation between Orbit Radius and Electron Velocity:

The radius r of the electron's orbit is related to its velocity v by the formula:

$$r = \frac{e^2}{4\pi\epsilon_0 mv^2}$$

4. Kinetic Energy (K):

The kinetic energy K of the electron is given by:

$$K = \frac{1}{2}mv^2 = \frac{e^2}{8\pi\epsilon_0 r}$$

5. Electrostatic Potential Energy (U):

The potential energy U of the electron due to the nucleus is:

$$U = -\frac{e^2}{4\pi\epsilon_0 r}$$

The negative sign indicates that the force is attractive.

6. Total Energy (E):

The total energy E of the electron in the hydrogen atom is the sum of its kinetic and potential energy:

$$E = K + U = \frac{e^2}{8\pi\epsilon_0 r} - \frac{e^2}{4\pi\epsilon_0 r}$$

$$E = -\frac{e^2}{8\pi\epsilon_0 r}$$

The total energy is negative, meaning the electron is bound to the nucleus.

Atomic Spectra:

1. Characteristic Spectrum of Elements:

Each element has its own unique spectrum of radiation, which it emits.

2. Emission Line Spectrum:

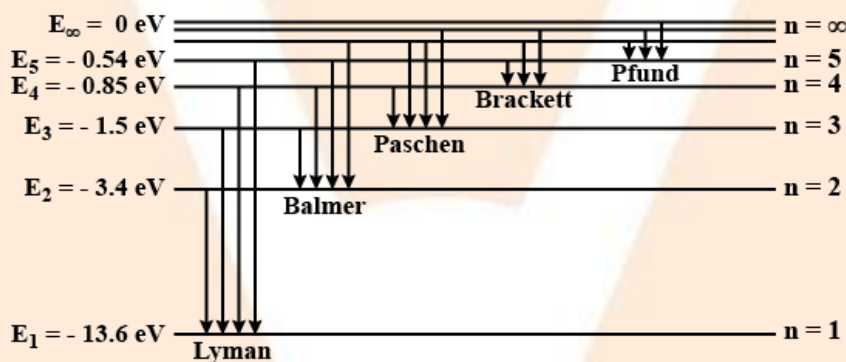
- When a gas or vapour is excited at low pressure (often by an electric current), it emits radiation with specific wavelengths.
- This is called the emission line spectrum, where bright lines appear on a dark background.
- The spectrum of atomic hydrogen is an example.

3. Identification through Emission Line Spectra:

The study of an element's emission line spectrum can be used like a "fingerprint" to identify the gas.

4. Absorption Spectrum:

- When white light passes through the gas, some wavelengths are absorbed, creating dark lines in the spectrum.
- These dark lines match the wavelengths in the emission line spectrum of the gas.
- This is called the absorption spectrum.



Energy level diagram for hydrogen atom

Bohr Atomic Model:

Bohr considered the Rutherford model of the atom and added a few arbitrary conditions in it. These conditions are referred to as his postulates:

I. The electron in a stable orbit does not radiate energy. i.e., $\frac{mv^2}{r} = \frac{kZe^2}{r^2}$

II. A stable orbit is that in which the angular momentum of the electron about nucleus is an integral (n) multiple of $\frac{h}{2\pi}$ i.e., $mvr = n \frac{h}{2\pi}$; $n = 1, 2, 3, \dots (n \neq 0)$

III. The electron can absorb or radiate energy only when the electron jumps from a lower to a higher orbit or falls from a higher to a lower orbit.

IV. The energy emitted or absorbed is a light photon of frequency ν and of energy, $E = h\nu$

10.1 For Hydrogen Atom:

($Z = \text{atomic number} = 1$): $L_n =$

1. angular momentum in the n^{th} orbit $= n \frac{h}{2\pi}$

2. $r_n =$ radius of n^{th} circular orbit $= (0.529 \text{ \AA}) n^2$

$$(1 \text{ \AA} = 10^{-10} \text{ m}) \Rightarrow r_n \propto n^2$$

3. $E_n =$ Energy of the electron in the n^{th} orbit $= \frac{-13.6 \text{ eV}}{n^2}$

$$\Rightarrow E_n \propto \frac{1}{n^2}$$

Note:

Total energy of the electron in an atom is negative, mentioning that it is bound.

Binding energy $(B.E)_n = -E_n = \frac{13.6 \text{ eV}}{n^2}$

4. $E_{n_2} - E_{n_1} =$ Energy emitted when an electron jumps from n_2^{th} orbit to n_1^{th} orbit ($n_2 > n_1$)

$$\Rightarrow \Delta E = (13.6 \text{ eV}) \left[\frac{1}{n_1^2} - \frac{1}{n_2^2} \right]$$

$$\Rightarrow \Delta E = h\nu$$

where,

ν = frequency of spectral line emitted.

$\frac{1}{\lambda} = \nu$ = wave number [number of waves in unit length]

$$\Rightarrow \nu = R \left[\frac{1}{n_1^2} - \frac{1}{n_2^2} \right]$$

where, R = Rydberg's constant for hydrogen = $1.097 \times 10^7 m^{-1}$

5. For hydrogen like atom/species of atomic number Z :

$$r_{nz} = \frac{\text{Bohr radius}}{Z} n^2 = (0.529 \text{ \AA}) \frac{n^2}{Z}$$

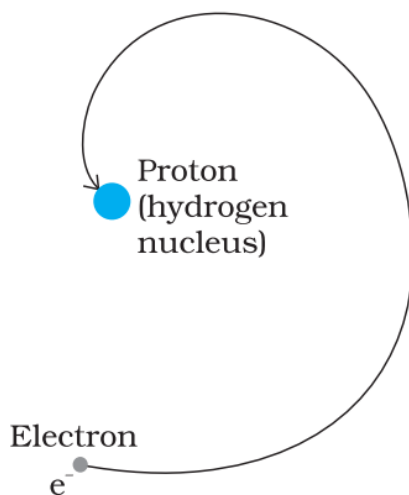
$$E_{nz} = (-13.6) \frac{Z^2}{n^2} \text{ eV}$$

$R_z = RZ^2$ – Rydberg's constant for the element of atomic number Z .

Note: When motion of the nucleus is also considered, then m is replaced by μ , where

$$\mu = \text{reduced mass of electron-nucleus system} = \frac{mM}{(m + M)}$$

In this case, $E_n = (-13.6 \text{ eV}) \frac{Z^2}{n^2} \frac{\mu}{m_e}$



Energy levels

1. Energy in Atom:

The energy of an atom is lowest (most negative) when the electron is closest to the nucleus. This happens when $n = 1$.

For higher values of n (like 2, 3, ...), the energy is larger, meaning the electron has more energy in outer orbits.

2. Ground State:

The lowest energy state, called the ground state, occurs when the electron is in the smallest orbit (Bohr radius, a_0 with energy $E_1 = -13.6 \text{ eV}$).

To remove the electron from this state (free it), the energy required is 13.6 eV, called ionisation energy of hydrogen.

3. Excited States:

- At room temperature, most hydrogen atoms are in their ground state.

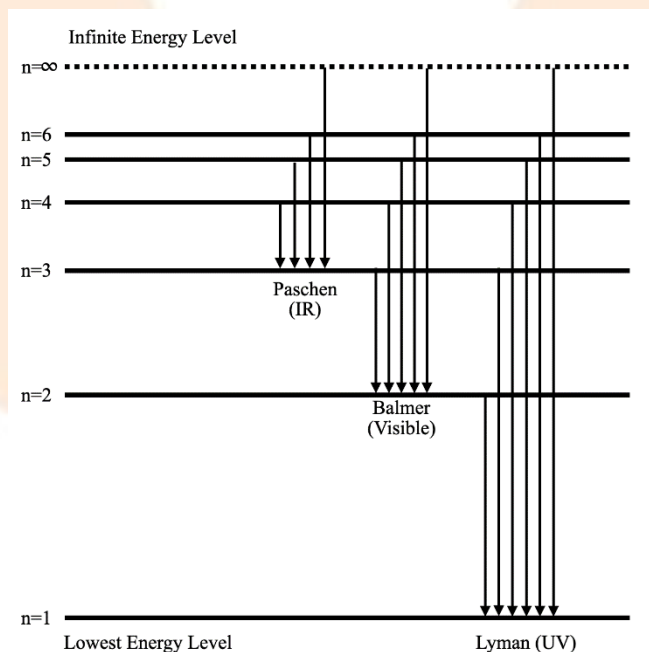
- When the atom gains energy (e.g., through collisions), the electron can jump to a higher energy level, entering an excited state.
- For $n = 2$, the energy is $E_2 = -3.40 \text{ eV}$. To move from $n = 1$ to $n = 2$, the energy required is 10.2 eV .

4. Higher Excited States:

- For $n = 3$, the energy is $E_3 = -1.51 \text{ eV}$, and moving from $n = 1$ to $n = 3$ requires 12.09 eV .
- As n increases, the energy levels become closer together, requiring less energy to move between them.

5. Energy Level Diagram:

- The highest energy state corresponds to $n = \infty$, where the energy is 0 eV , meaning the electron is completely free from the nucleus.
- As n increases, the excited state energies come closer together.



The Line Spectra of the Hydrogen Atom

1. Bohr's Third Postulate:

When an atom transitions from a higher energy state (n_i) to a lower energy state (n_f), the energy difference is released as a photon.

The frequency (ν_{if}) of the emitted photon is related to the energy difference:

$$h\nu_{if} = E_{n_i} - E_{n_f}$$

where h is Planck's constant.

2. Energy Levels of Hydrogen Atom:

The electron in a hydrogen atom usually stays in the ground state at room temperature.

To remove an electron from the ground state (ionisation), 13.6 eV of energy is needed.

Above $E = 0$, the electron is free, meaning there are infinite possible energy states.

3. Emission Lines:

When an electron moves from a higher to a lower energy level, it emits a photon. This creates spectral lines called emission lines.

4. Absorption Lines:

If an atom absorbs a photon with exactly the right energy, the electron moves to a higher energy state. This creates absorption lines in the spectrum.

5. Spectral Analysis:

When light passes through a gas, dark absorption lines appear in the spectrum at the frequencies absorbed by the atoms.

6. Bohr's Contribution:

Bohr's model of the hydrogen atom helped explain atomic spectra and was a major step toward quantum theory. He won the Nobel Prize in Physics in 1922 for this work.

Bohr's Second Postulate and de Broglie's Explanation:

1. Bohr's Second Postulate:

Bohr proposed that the angular momentum of an electron in orbit is quantized. The angular momentum can only take specific values:

$$L_n = \frac{nh}{2\pi}, \quad n = 1, 2, 3, \dots$$

2. Why is Angular Momentum Quantized?:

In 1923, Louis de Broglie explained this puzzle by proposing that electrons, like waves, have wave-like properties. This was 10 years after Bohr's model.

In Chapter 11, we learned that material particles (like electrons) have a wave nature, verified experimentally in 1927 by C.J. Davisson and L.H. Germer.

3. Wave Nature of Electrons:

de Broglie suggested that electrons moving in circular orbits should be viewed as particle waves, similar to standing waves on a string.

4. Standing Waves on a String:

Only specific wavelengths form standing waves when a string is plucked. Waves with different wavelengths interfere with themselves and disappear.

For an electron in the n th orbit of radius r_n , the total distance the wave travels is the orbit's circumference:

$$2\pi r_n = n\lambda, \quad n = 1, 2, 3, \dots$$

5. Electron Momentum and de Broglie Wavelength:

The de Broglie wavelength of an electron is given by:

$$\lambda = \frac{h}{p}$$

For an electron with speed v_n , the momentum is mv_n , so:

$$\lambda = \frac{h}{mv_n}$$

6. Bohr's Quantum Condition:

From the above equation, the quantum condition for angular momentum is derived as:

$$mv_n r_n = \frac{nh}{2\pi}$$

Limitations of Bohr's Model:

1. Applicable to Hydrogenic Atoms Only:

The Bohr model works for atoms with only one electron, like hydrogen, but not for multi-electron atoms like helium. This is because electrons also interact with each other, which Bohr's model doesn't account for.

2. Unable to Explain Relative Intensities:

The model correctly predicts the frequencies of light emitted by hydrogen atoms but cannot explain why some frequencies are stronger or weaker in intensity.

3. Quantum Mechanics Required:

For more complex atoms, a new theory called Quantum Mechanics is needed, which provides a more complete understanding of atomic structure.